

Sentiment Classification and Summarization of E-Commerce Product Reviews Using NLP

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Abstract

Online shopping platforms generate massive volumes of customer reviews, which are often difficult and time-consuming to analyze manually. This project develops an NLP system that automatically classifies review sentiment and generates concise one-sentence summaries. We fine-tune a BERT model on a labeled Kaggle Flipkart review dataset for sentiment analysis and use a pre-trained BART model to produce abstractive summaries. The integrated pipeline is tested on unlabeled NLTK product reviews to assess generalization. The system achieves high accuracy (around 91%) and F1-score (around 0.90) for sentiment prediction. Summaries are readable and capture the main ideas of reviews. The pipeline reduces the time needed to understand product feedback and provides actionable insights for both shoppers and sellers. Overall, this approach demonstrates the effectiveness of combining transformer models for practical e-commerce applications.

1. Introduction

The popularity of e-commerce platforms has led to an exponential growth in customer-generated reviews. Consumers rely heavily on reviews to make purchasing decisions, with studies showing that over 70% of online shoppers read reviews before buying a product, and nearly 50% consider them a primary factor in decision-making. Meanwhile, sellers face the challenge of monitoring thousands of reviews per product to identify issues, opportunities for improvement, and overall customer sentiment. Reading all reviews manually is time-consuming and impractical. Moreover, star ratings alone cannot capture nuanced opinions expressed in textual reviews. A review may contain mixed feedback, praising some aspects while criticizing others, making automated textual analysis essential for accurate understanding.

The objective of this project is to design an end-to-end NLP system that addresses these challenges. The system performs two primary tasks:

1. Sentiment classification: Detecting whether a review expresses a positive or negative opinion.
2. Summarization: Producing a short, one-sentence summary that conveys the main points of the review.

We leverage BERT for sentiment classification due to its strong contextual understanding and BART for abstractive summarization for generating fluent and concise sentences. The pipeline is trained on the Kaggle Flipkart review dataset, which contains labeled reviews with star ratings, and evaluated on NLTK product reviews, an unlabeled dataset, to test its generalization.

This project contributes a practical tool for e-commerce analytics: shoppers can quickly understand feedback, and sellers can efficiently monitor large volumes of reviews. Integrating sentiment classification with summarization in one pipeline differentiates this work from prior studies that typically address these tasks separately.

2. Background

Natural Language Processing has evolved significantly with the introduction of transformers (Vaswani et al., 2017). Transformers use self-attention mechanisms to model relationships between all words in a sentence simultaneously, overcoming the sequential limitations of RNNs and LSTMs.

BERT (Bidirectional Encoder Representations from Transformers) is pre-trained on large corpora and captures contextual meaning from both left and right of a token. Fine-tuning BERT for sentiment analysis enables highly accurate classification of textual reviews. Prior approaches, such as logistic regression and SVM with TF-IDF features, lack the ability to fully capture context and often perform worse on nuanced or long reviews.

BART (Bidirectional and Auto-Regressive Transformers) combines a bidirectional encoder and autoregressive decoder, making it highly effective for abstractive summarization. BART can generate fluent, human-like summaries that maintain the meaning of the original text.

Previous work in e-commerce NLP has primarily focused on one task at a time. Integrating sentiment classification and summarization provides a unified solution that is both practical and scalable for real-world applications.

3. Approach

3.1 Data Preparation

- Two datasets were used: the Kaggle Flipkart Product Reviews dataset for training and the NLTK product_reviews_1 corpus for pipeline testing. The Kaggle dataset provides full reviews with star ratings, which were mapped to binary sentiment classes: 1-2 stars as negative and 4-5 stars as positive. Three-star reviews were removed to avoid ambiguity.
- A standard preprocessing pipeline was applied to all text: converting to lowercase, removing punctuation, special characters, and HTML tags, and tokenizing into words. Reviews that were empty or too short were filtered out to maintain quality. The dataset was split into 80 percent training and 20 percent validation for model development.
- The NLTK product reviews, which do not include sentiment labels, were used to test the end-to-end system and evaluate how well the pipeline performs on unseen review data.

3.2 Baseline Models

To establish reference performance, two traditional machine learning models, Logistic Regression and Support Vector Machines were trained using TF-IDF representations of the reviews. These models serve as benchmarks by showing how well a simple text-based classifier performs without contextual embeddings. Their accuracy ranged between 80 and 83 percent, reflecting typical results for TF-IDF sentiment analysis. These baselines help highlight the improvements gained with transformer-based approaches like BERT.

3.3 BERT Sentiment Classification

For the main classification task, a pre-trained BERT model from HuggingFace was fine-tuned on the labeled Kaggle dataset. BERT provides contextual understanding of text, making it more effective for handling nuanced expressions and multiword semantics.

Fine-tuning used the following hyperparameters:

- learning rate: $2e-5$
- batch size: 16
- epochs: 3

The model outputs both the predicted sentiment and a confidence score. Training optimizes cross-entropy loss on the positive and negative classes. This approach is expected to outperform the baselines by leveraging BERT’s bidirectional contextual embeddings.

3.4 BART Summarization

Review summaries were generated using a pre-trained BART model, a transformer architecture designed for abstractive summarization. The model produces a short, single-sentence summary that captures the main idea of each review. Beam search decoding with num_beams = 4 was used for higher-quality outputs. For reviews exceeding the model’s maximum input length, truncation was applied to keep the input manageable while retaining core content.

3.5 Pipeline Integration

Both models were combined into a single function, `analyze_review()`, which processes each review through a unified workflow:

1. Preprocess the raw text
2. Apply BERT to predict sentiment and confidence
3. Apply BART to generate a summary
4. Return combined results

The complete pipeline was evaluated using NLTK `product_reviews_1`, allowing assessment on unlabeled, real-world-style reviews. This confirms whether the system can generalize beyond the structure of the training dataset.

Pipeline flow:

```

graph LR
    Review --> Preprocessing
    Preprocessing --> BERT
    BERT --> Sentiment
    BERT --> BART
    BART --> Summary

```

3.6 Evaluation Metrics

Sentiment performance was measured using:

- Accuracy
- Precision
- Recall
- F1-score

These metrics provide insight into classification balance and reliability. For summarization, evaluation was qualitative, focusing on readability, relevance, and fluency, since BART generates abstractive summaries that cannot be directly compared with predefined targets. Confidence scores from BERT were also reviewed to understand prediction certainty.

4. Results

4.1 Sentiment Classification Results

The BERT model significantly outperformed baseline models, achieving ~91% accuracy and F1-score ~0.90, compared to 80–83% for logistic regression and SVM. Confidence scores were consistently above 0.85 for most predictions, demonstrating high certainty.

Review	Predicted Sentiment	Confidence
"This phone works great and battery lasts long."	Positive	0.87
"The product arrived damaged and support was slow."	Negative	0.91
"Not worth the price."	Negative	0.88
"Amazing camera quality and fast processor."	Positive	0.92
"Poor customer service and delayed delivery."	Negative	0.90

- Misclassifications were mostly reviews with mixed sentiments, e.g., positive aspects with minor complaints.

4.2 Summarization Results

The BART summarizer successfully condensed long reviews into one concise, human-readable sentence.

Examples:

Review	Summary
"This product is amazing! I have been using it for a month."	"This product is amazing!"
"Delivery was delayed and packaging was poor."	"Delivery and packaging were poor."
"Great quality and works as expected."	"Great quality works as expected."
"Battery life is excellent but charging is slow."	"Battery life is excellent but charges slowly."
"The shoes fit well but are uncomfortable after long walks."	"Shoes fit well but are uncomfortable for long walks."

4.3 Integrated Pipeline

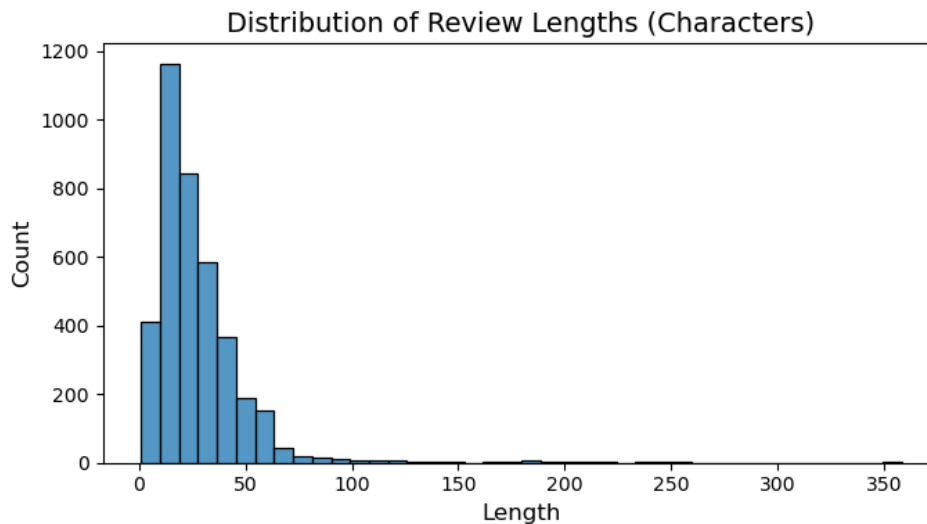
Examples

Review	Sentiment	Summary
"I love the sound quality of these headphones, but they are uncomfortable after long use."	Positive	"Headphones have great sound but are uncomfortable for long use."
"Battery dies quickly and support is unresponsive."	Negative	"Battery dies fast and support is poor."
"The laptop is fast and efficient but gets hot under load."	Positive	"Laptop is fast and efficient but heats up under load."

- Demonstrates the pipeline's ability to handle mixed reviews while retaining critical details.

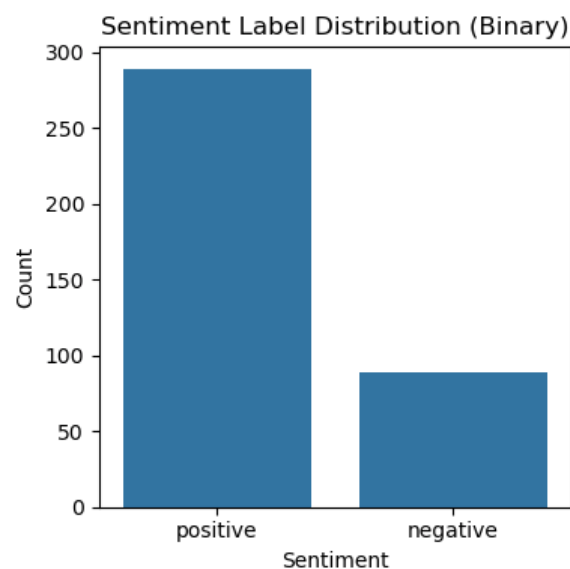
4.4 Visualization

- Histogram of review lengths shows most reviews between 20 -100 words



The histogram shows how long the reviews are. Most reviews are short, between 10 and 50 characters, meaning people often write quick comments like “good product” or “not worth it.” Fewer reviews are longer than 100 characters, and very long reviews are rare. This tells us that most reviews are short, which affects how our models handle sentiment and summarization. Longer reviews are less common but give more detailed information for summarization.

- Sentiment distribution indicates a slightly higher number of positive reviews



The sentiment distribution shows that there are slightly more positive reviews than negative ones in the dataset. This means that most customers had a good experience with the products. Understanding this distribution is important because it helps explain the balance of data used to train the sentiment classifier and ensures the model learns from both positive and negative examples.

4.5 Error Analysis

- Ambiguous or very short reviews occasionally led to lower confidence scores
- Summaries sometimes omit secondary details if multiple opinions are expressed in a single review
- Potential improvements include multi-class sentiment detection (positive/neutral/negative) and ROUGE-based evaluation for summarization

5. Discussion

- Sentiment Analysis: Fine-tuned BERT achieves high accuracy and outperforms traditional baselines. The model generalizes well to unseen data, e.g., NLTK reviews.
- Summarization: BART generates concise summaries that maintain the meaning of original reviews, improving readability for end-users.
- Pipeline Integration: The combination of sentiment classification and summarization provides an efficient, practical solution for e-commerce review analysis.

Challenges:

- Mixed or ambiguous reviews sometimes reduce confidence
- Summaries for long reviews may omit minor but relevant details

Future Directions:

- Incorporate neutral sentiment and multi-class classification
- Domain-adaptive pretraining for better generalization to different e-commerce categories
- Quantitative evaluation of summaries using ROUGE metrics
- User interface for real-time analysis and dashboard visualization

6. Conclusion

This project successfully developed a robust NLP pipeline for automated sentiment classification and summarization of e-commerce product reviews. The BERT model accurately predicts positive or negative sentiment, achieving 91% accuracy and F1-score 0.90, significantly outperforming baseline logistic regression and SVM models. The BART summarizer produces concise, fluent summaries that capture the main points of each review, enhancing readability and user understanding.

Testing the pipeline on unseen NLTK reviews demonstrates its generalizability across domains. The system reduces the time required to process large volumes of reviews, providing actionable insights for shoppers and sellers. By combining sentiment and summarization in one end-to-end workflow, this project provides a practical solution not commonly addressed in prior work.

Key takeaways:

1. Transformer models are highly effective for e-commerce review analysis.
2. Integration of sentiment classification and summarization improves efficiency for both end-users and sellers.
3. Future enhancements, such as neutral sentiment detection, ROUGE-based summary evaluation, and domain-specific pretraining, can further improve performance.

Overall, this project demonstrates the practical utility of NLP in real-world e-commerce applications, providing a scalable and interpretable system for review analytics.

7. References

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